🚀 Complete Setup Guide - Copy-Paste Ready Code

Step-by-Step Instructions for Beginners

📋 What You're Building

A professional stock analysis website with:

- Smart stock search
- Technical analysis
- News sentiment
- AI price predictions
- PDF reports
- Beautiful interface

📁 Step 1: Create Folder Structure

Create these folders on your computer:

```
1. Create main folder: "stock_analysis_system"
2. Inside it, create:
   - modules (folder)
   - models (folder)
   - assets (folder)
```

Your folder should look like this:

```
C:\stock_analysis_system\          (or wherever you want)
    ├── modules\
    ├── models\
    └── assets\
```

📄 Step 2: Create All Files

Create these files and copy-paste the code:

In main folder (stock_analysis_system):

1. `requirements.txt` → Copy from File 1
2. `app.py` → Copy from File 9

In modules folder:

1. `__init__.py` → Copy from File 2
2. `data_fetcher.py` → Copy from File 3
3. `technical_analysis.py` → Copy from File 4
4. `sentiment_analyzer.py` → Copy from File 5
5. `ml_predictor.py` → Copy from File 6
6. `visualization.py` → Copy from File 7
7. `pdf_generator.py` → Copy from File 8

Step 3: How to Create and Save Files

Windows:

1. Open Notepad
2. Copy-paste code
3. Click "File" → "Save As"
4. Select "All Files" (not .txt!)
5. Type exact filename (e.g., `app.py`)
6. Click Save

Mac:

1. Open TextEdit
2. Format → Make Plain Text
3. Copy-paste code
4. File → Save
5. Type exact filename (e.g., `app.py`)
6. Click Save

⚙ Step 4: Install Python Packages

Open Terminal/Command Prompt:

Windows:

1. Press Windows Key + R
2. Type `cmd`
3. Press Enter

Mac:

1. Press Command + Space
2. Type `terminal`
3. Press Enter

Navigate to your folder:

```
cd C:\stock_analysis_system
```

(Replace with your actual path)

Install packages:

```
pip install -r requirements.txt
```

This will take 5-10 minutes. Let it complete.

▯ Step 5: Run the Application

In the same terminal, type:

```
streamlit run app.py
```

What happens:

1. Terminal shows: "You can now view your Streamlit app in your browser"
2. Browser opens automatically
3. You see the stock analysis website!

If browser doesn't open automatically:

- Look in terminal for URL like: `http://localhost:8501`
- Copy and paste it in your browser

🎯 Step 6: Use the Application

Search for a Stock:

Method 1 - By Company Name:

1. Select "Company Name" in sidebar
2. Type: "Reliance" or "Bank of Baroda"
3. Click "Find Ticker"
4. System shows ticker symbol (e.g., RELIANCE.NS)

Method 2 - Direct Ticker:

1. Select "Ticker Symbol" in sidebar
2. Type: "RELIANCE.NS"
3. Ready to analyze

Analyze:

1. Click "ANALYZE STOCK" button
2. Wait 10-20 seconds
3. See complete analysis!

Generate Report:

1. After analysis, click "Generate PDF Report"
2. Click "Download Report"
3. Save to your computer

📄 Complete File Contents

FILE 1: requirements.txt

Save this in main folder as: **requirements.txt**

```
streamlit==1.29.0
pandas==2.1.0
numpy==1.25.0
yfinance==0.2.31
plotly==5.17.0
matplotlib==3.8.0
ta==0.11.0
xgboost==2.0.1
scikit-learn==1.3.2
shap==0.43.0
```

```
transformers==4.35.2
torch==2.1.1
sentencepiece==0.1.99
googlenews==1.6.10
requests==2.31.0
beautifulsoup4==4.12.2
weasyprint==60.1
jinja2==3.1.2
pillow==10.1.0
python-dotenv==1.0.0
joblib==1.3.2
tqdm==4.66.1
```

FILE 2: modules/init.py

Save in **modules** folder as: **init.py**

```
"""
Advanced Stock Analysis System - Core Modules
"""
__version__ = "2.0.0"
__author__ = "Stock Analysis Team"

__all__ = [
    'data_fetcher',
    'technical_analysis',
    'sentiment_analyzer',
    'ml_predictor',
    'visualization',
    'pdf_generator'
]
```

IMPORTANT NOTE:

For Files 3-9, please refer to the separate PDF documents I created earlier:

- **File 3** (data_fetcher.py) - See document "3-data-fetcher"
- **File 4** (technical_analysis.py) - See document "4-technical-analysis"
- **File 5** (sentiment_analyzer.py) - See document "5-sentiment-analyzer"
- **File 6** (ml_predictor.py) - See document "6-ml-predictor"
- **File 7** (visualization.py) - See document "7-visualization"
- **File 8** (pdf_generator.py) - See document "8-pdf-generator"
- **File 9** (app.py) - See document "9-Complete-App-Code"

Each file has extensive comments explaining every line of code!

🔧 Troubleshooting

Problem 1: "pip not recognized"

Solution: Python not installed or not in PATH

1. Download Python from: https://python.org
2. During installation, check "Add Python to PATH"
3. Restart computer
4. Try again

Problem 2: "streamlit not found"

Solution: Install failed

```
pip install streamlit --upgrade
```

Problem 3: "Module not found"

Solution: Wrong folder or files not saved correctly

1. Check you're in right folder: `cd C:\stock_analysis_system`
2. Check all files exist: `dir` (Windows) or `ls` (Mac)
3. Verify folder structure

Problem 4: "Ticker not found"

Solution: Add exchange suffix

Instead of "RELIANCE", use "RELIANCE.NS" (for NSE) or "RELIANCE.BO" (for BSE)

Problem 5: Application very slow

Solution: First time training model

- First analysis takes 2-5 minutes (training model)
- Next time same stock = much faster (uses saved model)
- This is normal!

✅ Checklist Before Running

- [ ] Created folder: stock_analysis_system
- [ ] Created subfolders: modules, models, assets
- [ ] Created requirements.txt in main folder

- [ ] Created app.py in main folder
- [ ] Created **init**.py in modules folder
- [ ] Created data_fetcher.py in modules folder
- [ ] Created technical_analysis.py in modules folder
- [ ] Created sentiment_analyzer.py in modules folder
- [ ] Created ml_predictor.py in modules folder
- [ ] Created visualization.py in modules folder
- [ ] Created pdf_generator.py in modules folder
- [ ] Installed packages: pip install -r requirements.txt
- [ ] All installations completed without errors

If all checked ✅, you're ready to run!

Understanding the Code

What Each File Does:

requirements.txt

- Lists all packages needed
- Like a shopping list for code

app.py

- Main application
- Creates the website
- This is what you run!

data_fetcher.py

- Gets stock prices from internet
- Searches for ticker symbols
- Downloads company information

technical_analysis.py

- Calculates indicators (RSI, MACD, etc.)
- Generates buy/sell signals
- Like a technical analyst

sentiment_analyzer.py

- Reads news articles
- Determines if news is positive/negative

- Calculates sentiment score

ml_predictor.py

- Uses AI to predict prices
- Trains machine learning model
- Saves/loads models

visualization.py

- Creates charts
- Interactive candlestick charts
- Indicator plots

pdf_generator.py

- Creates downloadable reports
- Formats analysis nicely
- Generates text reports

Tips for Beginners

1. Don't Modify Code Initially

- First, just copy-paste exactly as shown
- Get it working first
- Then experiment later

2. Read the Comments

- Every file has detailed comments
- Comments explain what each part does
- Start with "# This does..."

3. One Error at a Time

- If you get errors, fix one at a time
- Read error messages carefully
- Google the error if needed

4. Save Often

- Save files after copying code
- Check file names are exactly right
- No extra spaces in names!

5. Be Patient

- First run takes time
- Packages download takes 5-10 minutes
- First analysis takes 2-5 minutes
- After that, it's fast!

🆘 Getting Help

If Stuck:

1. **Check Folder Structure**
   - Use `dir` (Windows) or `ls` (Mac/Linux)
   - Make sure all files exist
2. **Check File Names**
   - Must be exactly: `app.py` (not `app.py.txt`)
   - Case sensitive: `__init__.py` (with two underscores)
3. **Read Error Messages**
   - Terminal shows what's wrong
   - Google the error message
   - Often easy fix!
4. **Reinstall Packages**

   ```
   pip uninstall streamlit
   pip install streamlit
   ```

🎉 Success!

When everything works, you'll see:

1. Terminal shows Streamlit starting
2. Browser opens automatically
3. Beautiful website loads

4. Sidebar with search box
5. Ready to analyze stocks!

First Test:

1. Type "Reliance" in company name
2. Click "Find Ticker"
3. Click "ANALYZE STOCK"
4. Wait for results
5. See complete analysis!

**Congratulations! You've built a professional stock analysis system! **

Next Steps

After Getting It Working:

1. **Try Different Stocks**
    - Bank of Baroda
    - TCS
    - Infosys
    - HDFC Bank
2. **Experiment with Settings**
    - Different time periods
    - Toggle indicators on/off
    - Generate PDF reports
3. **Learn the Code**
    - Read comments in each file
    - Understand what each part does
    - Try small modifications
4. **Enhance Features**
    - Add more indicators
    - Improve UI styling
    - Add more data sources

🎓 What You Learned

By building this, you learned:

- ✅ Python programming
- ✅ Streamlit web apps
- ✅ Stock market data
- ✅ Technical analysis
- ✅ Machine learning basics
- ✅ Data visualization
- ✅ Project structure

This is a professional-level project! 🎉

⚠ Important Notes

Data Disclaimer:

- Data from Yahoo Finance (free)
- May have 15-20 minute delay
- For educational purposes
- Not for actual trading decisions

Machine Learning:

- Predictions are not guarantees
- Past performance ≠ future results
- Always do your own research
- Consult financial advisors

Code License:

- Free to use for personal projects
- Free to modify and enhance
- Free to learn from
- Not for commercial use without permission

□ Bonus: Quick Commands Reference

Navigate to folder:

```
cd C:\stock_analysis_system
```

Install packages:

```
pip install -r requirements.txt
```

Run app:

```
streamlit run app.py
```

Stop app:

- Press `Ctrl + C` in terminal

Check Python version:

```
python --version
```

Check pip version:

```
pip --version
```

Update pip:

```
python -m pip install --upgrade pip
```

□ Final Words

You now have:

- ✅ Complete working code
- ✅ Detailed comments
- ✅ Setup instructions
- ✅ Troubleshooting guide
- ✅ Professional stock analysis system

Everything is copy-paste ready!

Just follow the steps, one by one, and you'll have a working application.

Good luck! You've got this!

Made with ❤ for beginners | Happy Coding!